



Assignment 2 CSC 225

Question 1

(a) loop 1 (n):

```
p ← 1
for i ← 1 to n2 - 1 do
    p ← p + i
```

lines:

- 1) p ← 1
- 2) for i ← 1 to n² - 1 do
- 3) p ← p + i

Line 1: 1A +

line 2: 1A + n² - 1 (1A + 1C) + 1C

line 3: n² - 1 (1A)

Total:

$$= 1A + 1A + n^2 - 1 (1A + 1C) + 1C + (n^2 - 1)(1A)$$

$$= 2 + 2n^2 - 2 + 1 + n^2 - 1$$

$$= 3n^2$$

(b) loop 2 (n):

```
line 1 S ← 1
line 2 for i ← 0 to n2 - 1 do
line 3     for j ← 0 to i do
line 4         S ← S * i
```

line 1: 1A

line 2: n² (1A + 1C)

line 3: $\frac{n^2(n^2+1)}{2} (1A + 1C)$

line 4: $\frac{n^2(n^2+1)}{2} (1A)$

Total Assignments: $2 + n^2 + n^2(n^2+1)$

Total Comparisons: $(n^2+1) + \frac{n^2(n^2+1)}{2}$

Question 2

$$T(n) = \begin{cases} 1, & \text{if } n = 1 \\ T(n-1) + n, & \text{if } n \geq 2 \end{cases}$$

Top down method:

$$\begin{aligned} T(n) &= T(n-1) + n \rightarrow T(n-1) = T((n-1)-1) + (n-1) \\ &= T(n-2) + n + (n-1) \rightarrow T(n-2) = T(n-3) + (n-2) \\ &= T(n-3) + n + (n-1) + (n-2) \end{aligned}$$

$\vdots$

$$= T(n-i) + n + (n-1) + (n-2) + \dots + (n - (i-1))$$

$$= T(n-i) + \sum_{i=n-i+1}^n i$$

$$= T(n-i) + \sum_{i=n-(n-1)+1}^n i$$

$$= T(1) + \sum_{i=2}^n i = 1 + \sum_{i=2}^n i$$

$$= \boxed{\sum_{i=1}^n i}$$

Question 3

$$T(n) = \begin{cases} 1, & \text{if } n = 0 \\ T(n-1) + 2^n, & \text{if } n \geq 1 \end{cases}$$

Show that $T(n) = 2^{n+1} - 1$ by induction

Base case: When $n = 0$, $T(0) = 2^{0+1} = 1 = 2^1 - 1 = 1$

So the base case holds.

I.H: Assume the formula holds true for some arbitrary $n = k$

$$\text{So, } T(k) = 2^{k+1} - 1$$

IS: To prove the formula holds true for $k+1$

$$T(k+1) = 2^{k+1+1} - 1 = 2^{k+2} - 1$$

So,

$$= T(k+1-1) + 2^{k+1}$$

$$= T(k) + 2^{k+1}$$

$$= 2^{k+1} - 1 + 2^{k+1}$$

$$= 2^{k+1+1} - 1 = 2^{k+2} - 1$$

Hence, by induction, the formula $T(n) = 2^{n+1} - 1$ holds true for all integers $n \geq 1$.

Question 4

arraySum(A, n)

loop invariant: At the start of the loop, for the given k , the sum is equal to the sum of elements.

$A[0]$ all the way until $A[k-1]$.

- Initialization: At the start of the first loop, the index k is initialized to 1. The variable sum is initialized to the 1st element of the array $A[0]$. The total sum of the element from $A[0]$ to $A[0]$ will be the element itself.
- Maintenance: During the k th iteration, the algorithm adds the current element $A[k]$ to sum.

$$\text{sum} = \text{sum} + A[k]$$

After this equation, sum becomes the sum of elements from $A[0]$ to $A[k]$

Thus, the loop invariant is maintained during each iteration of the loop:
: sum is the sum of elements $A[0]$ through $A[k]$

- Termination:

The loop terminates when $k=n$. The loop has iterated over all elements of the array. Since the loop invariant has been maintained through all iterations, when the loop finishes

Thus, when the loop terminates, the algorithm correctly returns the sum of the array A .

Question 5

Pseudocode [GCD]

Algorithm: GCD(a, b):

Input: Two integers a and b

Output: One integer, that is the greatest common divisor of the two integers a and b

if $b = 0$ then
 return a

else

 return GCD(b, a mod b)

end.

Recurrence equation

$$T(n) = \begin{cases} 2, & \text{if } n=0 \\ T(n-1)+2, & \text{if } n \geq 1 \end{cases}$$